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import time
from max30100 import MAX30100
sensor = MAX30100()
sensor.enable_spo2()

prev_ir = 0
last_beat_time = None
bpm_list = []
THRESHOLD = 3800
print("Measuring Heart Rate...")
while True:
    sensor.read_sensor()
    ir = sensor.ir
    if ir is None:
        continue if prev_ir > ir and prev_ir > THRESHOLD:
        current_time = time.time()

    if last_beat_time is not None:
        interval = current_time - last_beat_time bpm = 60 / interval

    if 45 < bpm < 160:
        bpm_list.append(bpm)
        avg_bpm = sum(bpm_list[-5:]) / len(bpm_list[-5:])
        print("Heart Rate:", int(avg_bpm), "BPM")
        last_beat_time = current_time
    prev_ir = ir
    time.sleep(0.03)
    if last_beat_time is not None:
        interval = current_time - last_beat_time bpm = 60 / interval

    if 45 < bpm < 160:
        bpm_list.append(bpm)
        avg_bpm = sum(bpm_list[-5:]) / len(bpm_list[-5:])
        print("Heart Rate:", int(avg_bpm), "BPM")

```